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Threat of entry and debt maturity: Evidence from airlines☆

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ABSTRACT

I explore the effect of the threat posed by low-cost competitors on debt structure in the airline industry. I use the route network expansion of low-cost airlines to identify routes where the probability of future entry increases dramatically. I find that when a large portion of their market is threatened, incumbents significantly increase debt maturity *before* entry occurs. Overall, the main findings suggest that airlines respond to entry threats trading off the benefits of short-term financing for lower rollover risk. The results are consistent with models in which firms set their optimal debt structure in the presence of costly rollover failure.

1. Introduction

Financial structure decisions have important implications for a firm's ability to survive in competitive markets. For instance, previous literature explains that "deep-pocketed firms" will attempt to drive financially constrained competitors out of business (see, e.g., Telser, 1966; Bolton and Scharfstein, 1990), while highly leveraged incumbents are less likely to survive in a competitive environment (Zingales, 1998). A natural ex ante implication of this is that if firms anticipate tougher competition in the future, they should seek to adapt their financial structure today.

How incumbents should accomplish that in practice is, however, unclear. Previous research indicates that leverage ratios are highly persistent (Lemmon et al., 2008), while holding excess cash reserves can be expensive (Jensen, 1986; Holmström and Tirole, 2000). A growing theoretical literature suggests that firms should avoid refinancing in difficult times, securing long-term financing just before bad news arrives and cash flows are affected (e.g.,

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Brunnermeier and Yogo, 2009). However, the interactions between financing choices and the expectations of future competition have generally been overlooked by the empirical literature.

This paper tests whether the threat of entry by low-cost competitors affects financing decisions using data from the American domestic airline industry. The choice of airlines as the main setting for such an analysis is driven by two considerations. First, domestic flights are (relatively) homogeneous products sold at low margins and, therefore, particularly exposed to low-cost competition. Second, while an increase in the threat of entry could have a trigger effect on financing decisions, it is empirically challenging to identify a setting in which such a threat is observable to the incumbent. A domestic airlines setting allows to exploit the predictable expansion of low-cost airlines as a method for identifying which markets are likely to be affected.

Building on Goolsbee and Syverson (2008), I use the evolution in the expansion of route networks by major, low-cost carriers to identify which markets are threatened. In particular, I use domestic flight data collected by the U.S. Department of Transportation from 1990 to 2014 to establish flight routes for each air carrier. The focus is on situations where a low-cost carrier begins to operate at one endpoint of a route (having already been operating out of the other endpoint), *but before it starts flying the route connecting the two endpoints*.

As an illustrative example, consider Southwest's entry into Washington Dulles International Airport. Southwest began to fly out of Dulles (IAD) in October 2006. However, upon entering Dulles airport, Southwest did not immediately start flying on the route Dulles (IAD)-Cleveland (CLE). Cleveland is also a Southwest airport: The airline flew between CLE and other airports, but did not offer any CLE-IAD connections at first. It is however reasonable to expect that, after Southwest began to operate at both endpoints of the route, competing airlines soon realized that the probability of Southwest entering the Dulles (IAD)-Cleveland (CLE) route had risen dramatically (in fact, it started to fly the route in 2007).[1]

In my empirical analysis, I show that conditional on entry at the second endpoint airport of a route in year t, the probability of actual entry in the route in the following year increases by 25 percentage points with respect to an unconditional probability of less than 2%. Similarly, the probability of entry within the next three years increases by 36 percentage points.[2] Aggregating route-level threats, weighted by the passenger traffic on each route, I construct an *airline-level* measure of threat of entry, measuring for each airline which fraction of its market is exposed to the likely entry of low-cost competitors. Using a 10% random sample of tickets sold, I show that the actual entry of low-cost airlines has a disruptive effect on prices: average fares charged on the route drop by 7.3% against an average profit margin in the airline industry of 1% only.[3] Such a dramatic impact suggests that threatened airlines should seek to improve their financial structure before entry occurs.

This paper finds that a one standard deviation increase in the threat of entry is associated with an increase of 6.0 percentage points in the proportion of long-term debt over total debt held by incumbent airlines (a 10% increase relative to the baseline of 60%). This effect is particularly strong for airlines whose debt is rated as "speculative" and that are financially constrained—airlines that in general have more difficult access to credit. To provide additional support to my findings, I build a data set of debt issuances and explore how the threat of entry relates to debt characteristics. I find that threatened airlines issue debt instruments with longer maturity. Additionally, I show that while airlines are in general more likely to issue bonds, threatened airlines extend their debt maturity mostly via loans.

Furthermore, I show that the threat of entry by low-cost competitors has no significant effect on leverage. While threatened airlines would probably benefit from having lower debt, the air carriers in my sample present a highly persistent capital structure, which is unlikely to be adjusted annually (for instance, because it depends on the physical assets in place or follows tight debt to equity targets). Conversely, refinancing cycles are usually relatively short, allowing firms to adjust their debt composition more frequently.

Overall, my results are consistent with models deriving the optimal debt maturity structure in the presence of costly rollover failure. Longer debt maturity allows firms to reduce rollover (or liquidity) risk—the risk that lenders are unwilling to refinance when bad news arrives. Rollover risk increases credit risk (He and Xiong, 2012; Nagler, 2015), magnifies the debt overhang problem (Diamond and He, 2014), weakens investment (Almeida et al., 2012), and exposes the firm to the risk of costly debt restructuring (Brunnermeier and Yogo, 2009). My findings suggest that airlines find it optimal to commit to longer debt maturity to minimize the cost of rollover risk when the entry of low-cost competitors is likely. I provide evidence against alternative explanations based on shifts in the demand for transportation, changes in financial market conditions, agency costs, or mechanical adjustments.

This paper contributes to three streams of literature. First, a growing literature has been devoted to exploring the implications of rollover risk in different settings. Rollover risk played a role in both the default of Bear Stearns and Lehman Brothers as well as the general decrease in corporate investment in the United States following the financial crisis and ensuing credit crunch. Graham and Harvey (2001) indicate, using a survey of 392 financial executives, that the cost of refinancing in "bad times" is the second most important factor affecting the decision to issue long-term debt. Several recent theoretical papers predict that firms should manage rollover risk when

[1] This example is quoted from Goolsbee and Syverson (2008).

[2] Estimates for Southwest Airlines are reported. Different estimates are obtained for different low-cost carriers. The lowest coefficients are found for AirTran (12 and 18 percentage points for entry in the next year and within the next three years, respectively). The highest coefficients are found for Virgin America (34 and 65 percentage points).

[3] The number is from the International Air Transport Association (IATA) annual report available at http://www.iata.org/about/documents/iata-annual-review-2013-en.pdf.

they expect a negative shock to occur with high probability (Diamond, 1991a; Brunnermeier and Yogo, 2009; Diamond and He, 2014). This paper provides a first attempt at offering empirical support for that prediction.

Second, an extensive literature analyzes the determinants of capital structure. Most existing studies focus on the choice between equity and debt, treating the latter as uniform. However, a growing body of research recognizes that there is wide heterogeneity in debt and that firms might adjust their debt structure, which is more flexible, while leaving debt ratios unchanged (Rauh and Sufi, 2010; Colla et al., 2013). My paper adds to those studies, showing that threatened firms increase debt maturity without significantly changing their level of leverage. In particular, previous papers document that debt maturity is affected by a firm's quality (Guedes and Opler, 1996), agency costs (Barclay and Smith, 1995), firm size and asset maturity (Stohs and Mauer, 1996), information asymmetries (Berger et al., 2005), growth opportunities (Billett et al., 2007), asset liquidation values (Benmelech, 2009), and changes in the supply of long-term government bonds (Greenwood et al., 2010; Badoer and James, 2016). My results suggest that the threat of entry is an additional economically relevant determinant of the structure of debt maturity.

Finally, a growing literature in finance explores empirically the relation between product market competition and corporate variables. A series of papers explores the contemporaneous relation between competition and corporate policies (e.g., Chevalier, 1995; Zingales, 1998; Khanna and Tice, 2000; Giroud and Mueller, 2011; Xu, 2012; and Valta, 2012). Another body of literature focuses instead on the implications of competitive threats for corporate decisions (e.g., Goolsbee and Syverson, 2008; Hoberg et al., 2014; Frésard and Valta, 2016; and Cookson, 2017; 2018). Related to the latter stream of papers, my analysis focuses on the role of expectations before an actual entry decision is made. To the best of my knowledge this is the first paper to explore the effect of competitive threats by low-cost competitors on debt maturity structure. To study how firms adjust debt composition conditional on their expectations appears particularly important in light of the increasing evidence on the costs of rollover failure.

The remainder of the paper is organized as follows. Section 2 presents the testable hypotheses, Section 3 describes the data used, Section 4 briefly outlines the empirical design, Section 5 presents the main results, and Section 6 discusses the effects on debt composition. Section 7 concludes.

2. Hypotheses

In a frictionless world, firms should always be able to finance positive net present value (NPV) projects. However, in the presence of frictions, efficient firms could be forced to exit the market due to a lack of funds. Do firms take this into account by adapting their debt maturity and/or capital structure in anticipation of a negative shock? In the following, I develop my working hypotheses against the null of no change in financial structure in response to entry threats.

Hypothesis I. Airlines respond to entry threats by extending debt maturity.

The entry of a disruptive competitor affects the survival chances of the incumbent, potentially compromising its access to external funding. This, in turn, may force the incumbent to face inefficient rollover failure costs, especially when entry coincides with a refinancing moment. For example, these costs may entail having to seek more expensive sources of financing, renegotiating the terms of the debt contract with dispersed lenders (Buchheit and Gultai, 2004), or liquidating assets at fire-sale prices (Pulvino, 1998; Shleifer and Vishny, 2011).

Debt maturity decisions can affect the likelihood of a rollover failure. On the one hand, there are a number of benefits associated with borrowing at short maturities. For instance, short-term debt increases a firm's financial flexibility (Brunnermeier and Yogo, 2009), signals high-quality projects to the lenders (Flannery, 1986), reduces agency costs (Myers, 1977), and is preferred by creditors (Brunnermeier and Oehmke, 2013). On the other hand, a shorter debt maturity increases rollover risk, as it forces the firm to refinance frequently. When frequent refinancing leads to costly rollover failures, the incumbent should trade off the benefits of short-term financing for lower rollover risk. An optimal financing strategy would, therefore, require an incumbent to issue debt of the shortest maturity as long as subsequent rollover is guaranteed and debt of longer maturity just before rollover failure could occur (see Brunnermeier and Yogo, 2009).

Hypothesis II. Airlines respond to entry threats by decreasing leverage.

As an alternative strategy, incumbents could decrease leverage either by issuing equity or by reducing the amount of outstanding debt. Previous papers indicate that debt decreases a firm's flexibility, making it more vulnerable to competition. As a consequence, highly in debt firms are more easily pushed out of the market and are less likely to enter into a price war or to expand (see, e.g., Bolton and Scharfstein, 1990; Chevalier, 1995; Zingales, 1998; Khanna and Tice, 2000). This suggests that it might be optimal from an incumbent perspective to decrease leverage ex ante, before actual entry occurs.

There are, however, a number of considerations that suggest that incumbents should find it easier to adjust debt maturity rather than leverage. First, issuance costs deter most firms from frequently optimizing their capital structure (see, e.g., Strebulaev, 2007; Danis et al., 2014). According to the pecking order theory, these costs are going to be higher for equity issuance, thereby suggesting that a firm should find it relatively cheaper to issue debt. Second, the choice of issuing equity to reduce leverage could actually result in a net transfer of value from the shareholders to the lenders, in the event of a default triggered by the entry of a competitor (as debt is senior to equity). This makes issuing equity even more expensive ex ante (He and Xiong, 2012). Third, some firms target tight debt to equity ratios, whereas maturity targets are uncommon (Graham and Harvey, 2001). Fourth, it usually takes time to issue new debt or equity for a special purpose, and responding to entry threats is time sensitive. In this regard, renegotiating the debt maturity of the outstanding debt is arguably faster. Finally, as an alternative strategy to reduce leverage, the incumbent could deplete its cash reserves to buy back the outstanding debt. This would, however, leave it particularly exposed to price wars when the low-cost competitor eventually enters the market (Bolton and Scharfstein, 1990; Frésard, 2010). In contrast, every year, part of a firm's outstanding debt reaches maturity and new debt needs to be issued to replace it, thereby making its debt maturity structure significantly more flexible.[4]

Hypothesis III. Airlines respond to entry threats when highly exposed to rollover risk.

Different exposure to rollover risk could induce heterogeneity in the response to entry threats. According to Diamond (1991a), low-quality firms should actively seek to obtain long-term financing before entry occurs. Conversely, high-quality firms should always borrow at short maturities, as they have lower or no exposure to rollover risk. Importantly, the worst firms (e.g., those in distress) will not be able to borrow at long maturities, as the lenders will want to retain the right to liquidate them early in the event of a disruptive entry (Diamond, 1991a). However, I do not focus on this last category of firms in my empirical analysis, as the number of distressed firms in my sample is limited.

Another important element to consider is that the setting of this paper presents a difference with the standard rollover risk framework. Specifically, many recent studies focus on the costs of refinancing during temporary downturns, when credit markets are tight. Conversely, the entry of low-cost competitors constitutes a relatively persistent shock. In the presence of a permanently tougher environment, eliminating rollover risk completely would require a firm to issue debt that matches perfectly the maturity of its assets. However, the range of available maturities is constrained by lenders' preference for short-term debt (see, e.g., Brunnermeier and Oehmke, 2013; Custódio et al., 2013). In equilibrium, a threatened incumbent will therefore issue at the longest maturity that the lenders are willing to finance (Diamond, 1991a). This minimizes the number of refinancing episodes, thereby reducing the number of potential rollover failures (see Section 6.1 for a more detailed discussion on the supply side).

3. Data and summary statistics

To conduct my analysis, I use data from several different sources. US Department of Transportation Form 41 filings contain financial information on public and private airlines operating in the United States from 1990 to 2014. Compustat North America Fundamentals Annual files include all necessary public firm financials. Compustat Industry Specific Annual includes specific data on the airline industry. The T-100 Domestic Market data set contains data on routes and passenger traffic. Mergent includes data on bond issuance, while Dealscan contains data on loans. For most of the main analysis, I use data obtained by matching financial data from Compustat North America Fundamentals Annual and the T-100 Domestic Market data set according to airline names.

3.1. Airline data

I have two main available sources of financial data for airlines: Compustat North America Fundamentals Annual and compulsory filings of all airlines operating domestic flights in the United States (commonly referred to as Form 41) that I obtain from the U.S. Department of Transportation. I match airlines' flights to other Form 41 data sets using the variable "airline ID." Importantly, both samples are free from selection bias as the U.S. Department of Transportation makes data available for all operating and defunct airlines for the 1990–2014 period. I match airlines from the Form 41 data set to corresponding entries in Compustat financials by name. This significantly reduces the sample size, because most of the airlines operating in the United States are private regional airlines that are not included in Compustat. However, the Form 41 data have significantly less detailed information for such private carriers and are missing several control variables.[5] Therefore, I present results using Compustat in Section 5 and results from Form 41 filings (unmatched to Compustat) as corroborating evidence in the Online Appendix.

The Form 41 sample includes 140 airlines. This number falls down to 30 passenger airlines and a total of 384 observations after I match it to Compustat. Those airlines, however, cover more than 80% of the total domestic passenger traffic. To conduct my analysis, I focus on the smaller sample for which all key financial variables are available (i.e., *Q, Maturity, Profitability, Tangibility, Asset maturity*, and *Sales*). This leaves me with 239 observations. The effects of this selection are discussed in the Online Appendix. All results presented in the paper also hold in the larger sample.

In my analysis, I consider as low-cost airlines Southwest Airlines, JetBlue, Allegiant Air, Frontier Airlines, AirTran, and Virgin America. AirTran is excluded from the sample after it was integrated into Southwest.[6] In the Online Appendix, results are listed after dropping one low-cost airline at the time from the calculation of *Threat of entry* to make sure results are not driven by a single low-cost airline.

[4] Consistent with the arguments above, Rauh and Sufi (2010) find that a large fraction of firms that experience no significant one-year change in leverage, significantly adjust the underlying debt composition.

[5] In particular, the maturity of long-term debt is not specified in Form 41. Therefore, I can only distinguish between long-term debt due in one year and more than one year.

[6] Codeshare agreements and alliances could potentially bias my results because incumbents may choose not to alter their debt maturity structure when the threat of entry arises from a "friendly" airline. However, such alliances are rare for low-cost airlines. Southwest entered into a codeshare agreement with AirTran in 2013. JetBlue has several codeshare agreements with international carriers, but none of those carriers is included in my sample. Allegiant has no alliances or agreements with other companies. Frontier has a codeshare agreement with Great Lakes Airlines. However, dropping the "connected airlines" does not significantly alter the results.

3.2. Flight data

Data on flights are obtained from the T-100 Domestic *Market* data set collected by the U.S. Department of Transportation. These data have an important conceptual difference with the T-100 Domestic *Segment* data set. The former considers a route to be a "market" on the basis of its origin and destination city, no matter how many stops occur in between. The latter assumes that every stop breaks the flight into different markets: e.g., passengers taking off from Boston Logan (BOS) for destination Santa Barbara (SBA) with one stopover in Phoenix (PHX) and passengers flying from Boston (BOS) to Santa Barbara (SBA) without any stopover (or with a different stopover) are considered as buying the same product by the T-100 Domestic Market database, and buying two completely different products by the segment database. In the paper, I present results based on markets.

Another important distinction is between airports and cities. Computing routes on the basis of airports assumes that two flights taking off from the same airport but landing in two different airports of the *same* city operate in completely different markets. Conversely, computing routes on the basis of cities assumes that travelers are indifferent between airports located in the same city. Low-cost airlines often do not operate in the main airport of a city but in a less busy one. For instance, Southwest Airlines does not fly from Chicago O'Hare, which is the main Chicago airport and one of the busiest airports in the world by number of takeoffs and landings. Instead, Southwest operates in Chicago Midway, a smaller airport situated eight miles from Chicago's downtown. Therefore, in my analysis I determine routes on the basis of cities and not airports. For instance, I assume that the route from the Logan Airport in Boston to Chicago O'Hare would be affected if Southwest starts flying from Boston Logan to Chicago Midway. My flight sample is complete in the sense that every single domestic flight that took off in the 1990–2014 period is recorded. The matching of flight data with airlines' financials is conducted by airline name as indicated above.

3.3. Financial variables and summary statistics

The main variable of interest considered in my analysis is *Debt maturity*. I define *Debt maturity* as the percentage of debt maturing in more than three years following the literature (see, e.g., Barclay and Smith, 1995; Billett et al., 2007; Custódio et al., 2013; and Harford et al., 2014). An alternative approach would be to compute the weighted average maturity of all outstanding debt instruments (see, e.g., Guedes and Opler, 1996; and Benmelech, 2009). However, the former measure captures a firm's exposure to rollover risk while the latter does it only partially. For instance, a low value of the former measure signals that a firm will have to go through major debt rollover within the next three years. Conversely, the latter measure has less clear implications. A firm that holds debt instruments with *average* maturity of 12 years might actually have higher exposure to rollover risk than a firm holding debt instruments with average maturity of ten years if a larger part of the debt instruments of the former needs to be rolled over at the same time. Additionally, the latter measure is difficult to construct with my data.

As alternative measures of debt maturity, I present results using *Maturity*5, i.e., the percentage of debt maturing in more than five years, and *Long – term debt issuance*, i.e., the proportion of long-term debt that has just been issued (scaled by book assets). All variables constructed from accounting quantities are winsorized at the 1st and 99th percentiles to attenuate the effect of misreporting and outliers.

For the average airline in my sample, 59% of the debt matures in more than three years: this number is 67% for the median airline (see Table 1, Panel A). The construction of financial variables is described in Table A.1 in Appendix A. Variables based on flight data—like *Threat of entry* and *Entry*—are not winsorized as extreme situations are of particular interest, and the variables are already bounded by construction to be between zero and one. Airlines are on average bigger and more leveraged than other firms in Compustat. The average book leverage is around 36%, a number about twice that reported by related studies on manufacturing firms; see, e.g., Xu (2012). Airlines in my sample hold on average 14% in cash and exhibit a large share of tangible assets 55% versus 25% in the manufacturing industry; see Xu (2012).

On average 16% of the incumbents' market is under the threat of entry by low-cost carriers. This number, however, changes significantly from one airline to another and from year to year. Almost 25% of the airline-year pairs in my sample face a negligible threat of entry of less than 4% of their market, while for another 25% of the airline-year pairs more than 23% of the total market is threatened. The distribution of the other financial variables is described in Panel A of Table 1. Panel B of Table 1 reports data on fleets: the average number of aircraft operated by the airlines in my sample is 279. Furthermore, the average airline is able to fill 76% of the available seats (load factor), leases more than half of its fleet, and pays $2.05 per gallon of fuel. Panel C presents data on debt issues: the average maturity of debt issues in my sample is 135 months, 62% of debt issues have covenants, 74% are bonds, and 61% of the bonds are asset-backed.

In Table 1 Panel D, I report pairwise correlations between *Threat of entry, Debt maturity*, and other main financial variables. *Debt maturity* and *Threat of entry* display a correlation of 13%. Importantly, *Threat of entry* exhibits a correlation below 10% with all the other financial variables at the incumbent level (with the only exception being *Asset maturity*).[7] This partially mitigates the concern that *Threat of entry* might be affected by the characteristics of the incumbent. Consistent with related studies based on all industries, debt maturity in my sample displays a positive correlation with *Log sales* (Barclay and Smith, 1995), *Tangibility* (Benmelech, 2009), and *Asset maturity* (Guedes and Opler, 1996), and a negative correlation with Tobin's Q (Barclay and Smith, 1995; Guedes and Opler, 1996).

[7] To account for the correlation between *Asset maturity* and *Threat of entry*, I include *Asset maturity* as a control in all my regressions.

Table 1
Summary statistics.
This table presents summary statistics and pairwise correlations among the main variables. Panel A includes observations obtained by matching Compustat to flight route data (T-100 Domestic Market data set), Panel B shows fleet statistics obtained by matching Compustat Financials, Compustat Industry Specific Annual, and flight data from the T-100 Domestic Market data set. Panel C includes data obtained by matching data on debt issuances from Mergent and Dealscan to the main data set, the reported statistics are at the issuance level. Panel D shows pairwise correlations among the main variables. The sample period goes from 1991 to 2014. All variables are defined in Table A.1 in Appendix A.

Panel A: Airline characteristics						
	Obs.	Mean	25th	Median	75th	SD
Threat of entry	239	0.1608	0.0377	0.1096	0.2302	0.1706
Entry	239	0.0818	0.0195	0.0640	0.1155	0.0823
Maturity	239	0.5934	0.5219	0.6679	0.7697	0.2633
Issuance	239	0.0780	0.0081	0.0508	0.1033	0.0985
Leverage	239	0.3642	0.2102	0.3859	0.5086	0.1889
Cash holdings	239	0.1368	0.0670	0.1015	0.1946	0.0991
Tangibility	239	0.5457	0.4252	0.5765	0.6820	0.1887
Profitability	239	0.0816	0.0396	0.0940	0.1418	0.1492
Chapter 11	239	0.0418	0.0000	0.0000	0.0000	0.2006
Investment grade	239	0.1255	0.0000	0.0000	0.0000	0.3320
Q	239	1.3857	1.0377	1.1948	1.5584	0.5675
ΔPassengers	239	0.1489	−0.0032	0.0472	0.1361	0.4589
Asset maturity	239	9.1712	4.4060	9.0852	12.5105	5.5429
Investment	239	0.1029	0.0368	0.0781	0.1480	0.0871
Market share	239	5.9358	4.6613	5.8661	6.7881	1.5839
Panel B: Fleet characteristics						
Number of aircraft	175	278.5250	57.0000	193.0000	410.0000	246.2140
Age aircraft (in years)	122	10.2408	6.5000	9.4000	12.4900	5.4416
Leased aircraft/Total aircraft	161	0.5291	0.2972	0.4528	0.7489	0.2870
Fuel price p/gallon (in cents)	152	205.1063	97.1950	212.0000	310.0000	101.2040
Load factor (in %)	175	75.8742	72.4000	77.3000	81.6000	8.7621
Operating expenses p/seat-mile	174	11.3540	9.0300	10.3600	12.1200	5.1204
Avg. ticket price (in $)	157	115.8208	83.9821	106.6022	140.4772	43.4944
Panel C: Issuance characteristics						
Issue maturity (in months)	445	135	60	121	202	88
Covenants	445	0.62	0.0000	1.0000	1.0000	0.4864
Asset backed	330	0.61	0.0000	1.0000	1.0000	0.4886
Bond	445	0.74	0.0000	1.0000	1.0000	0.4382
Panel D: Correlation coefficients						
	Maturity	Threat	Sales	Tang	Asset maturity	Q
Maturity	1					
Threat of entry	0.1298	1				
Sales	0.3962	0.0687	1			
Tangibility	0.4375	0.0790	0.2670	1		
Asset maturity	0.4966	0.1569	0.2408	0.7540	1	
Q	−0.2190	0.0332	−0.2921	−0.2022	−0.1664	1

4. Empirical design

4.1. Identification strategy

This paper joins a recent stream of literature that attempts to measure exposure to competitive threats at the firm level (see, e.g., Cookson, 2017, 2018; Hoberg et al., 2014). Fig. 1 shows that, at the beginning of the 1990s, mostly airlines flying on routes in the South-West were exposed to the competition of low-cost carriers, while competition on the North-East routes was low. This situation reverses at the end of the following decade (see Fig. 2). To assume that all carriers are exposed to the same degree of competition at a certain time would therefore be highly misleading. In particular, I exploit the result, provided in Goolsbee and Syverson (2008), that Southwest's airport presence is a strong predictor of actual route entry.[8] Specifically, when Southwest enters the second endpoint airport of a route but not the route itself, the probability that it will enter the route "soon" increases dramatically (see Fig. 3). In this paper, I first generalize this approach to the six major low-cost airlines in the United States. Second, I aggregate this measure at the airline level to estimate the overall exposure of each incumbent airline's network to the future entry of low-cost competitors.[9]

[8] Empirical work showing that endpoint airport presence is correlated with entry includes Berry (1992) and Peteraf and Reed (1994), while Bailey (1981) describes a case in which the airport presence is used in antitrust policy. More broadly, the importance of airport presence is stressed in Borenstein (1989, 1990).

[9] The airline industry is often used as a laboratory to address broader questions due to its large data availability. For instance, Borenstein (1990) and Kim and Singal (1993) study the effect of airline mergers on

Fig. 1. Airport presence of low-cost carriers at the end of 1990. Low-cost carriers are Southwest Airlines, JetBlue, Allegiant Airlines, Frontier Airlines, AirTran, and Virgin Airlines. Data are obtained from the T-100 Domestic Market database.

Fig. 2. Airport presence of low-cost carriers at the end of 2014. Low-cost carriers are Southwest Airlines, JetBlue, Allegiant Airlines, Frontier Airlines, AirTran, and Virgin Airlines. Data are obtained from the T-100 Domestic Market database.

In my analysis, I focus on the threat posed by low-cost airlines as opposed to legacy (i.e., non-low cost) carriers for three reasons. First, using a 10% random sample of tickets sold by domestic airlines, I find that when a low-cost

market power; Borenstein and Rose (1995) explore the pricing implications of airline bankruptcy; Forbes and Lederman (2009) look at the effect of ownership on renegotiation costs; Forbes and Lederman (2010) investigate the effect of vertical integration on operational performance in the airline industry; Borenstein and Rose (1994) study ticket price discrimination; Benmelech and Bergman (2009) explore the effects of collateral value on the cost of debt; Benmelech and Bergman (2011) study the negative externalities of bankruptcy; and Azar et al. (2018) explore the effect of common ownership on competition.

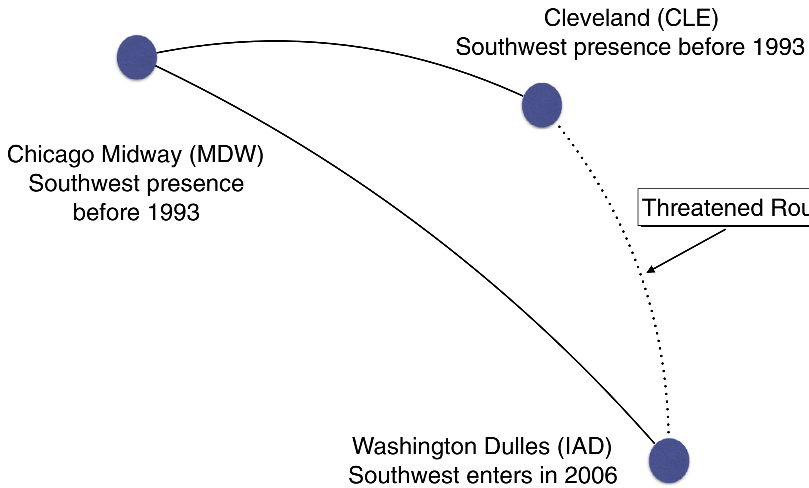

Fig. 3. When a low-cost carrier enters the second endpoint airport of a route *but not yet on the route itself* the route is considered under threat since the probability of entry in the following year increases dramatically. For instance, Southwest started to fly from Dulles (IAD) in October 2006. However, upon entering Dulles airport, Southwest did not immediately start flying on the route Dulles (IAD)-Cleveland (CLE). Cleveland is a Southwest airport: the airline flew between CLE and other airports, just not the CLE-IAD route. It is therefore reasonable to expect that, after Southwest began to operate at both endpoints of the route, competing airlines soon realized that the probability of Southwest entering the Dulles (IAD)-Cleveland (CLE) route had risen dramatically (see Goolsbee and Syverson, 2008).

airline starts flying on a route, average fares charged on the route drop by 7.3%. This number is particularly striking if one considers that the average profit margin in the airline industry is around 1% (see, e.g., the 2013 IATA annual report). Conversely, the entry of legacy carriers has a much more limited effect on prices. Second, low-cost airlines have expanded significantly in the last ten to 15 years, suggesting that all existing airlines had to face low-cost competition eventually. Third, alliances and codeshare agreements are very common among legacy carriers as a way to increase efficiency or respond to competitive pressure (Goetz and Shapiro, 2012). On the contrary, low-cost airlines generally avoid such agreements and, as a consequence, are mostly regarded as competitors by other airlines operating in the same market.

To identify which routes are under the threat of entry, I exploit that low-cost airlines do not expand randomly but are more likely to enter into a route if they are already present at both endpoint airports. Therefore, the entry in year t at the second endpoint airport increases disproportionally the probability of actual entry on the route (usually in year t+1 or t+2). I run probit regressions for the probability of a low-cost carrier's actual entry into a route in year $t+1$, conditional on its entrance at the second endpoint airport of the route in year t. The marginal probabilities are reported in Table 2 for entry in years t+1, t+2, and t+3 (time fixed effects are included and errors are clustered at the route level). Estimates for the marginal probabilities of entry in year t+1 conditional on entry at the second endpoint of a route (but not on the route itself) in year t are: 25% (Southwest), 17% (JetBlue), 17% (Allegiant), 26% (Frontier), 12% (AirTran), and 33% (Virgin America). All coefficients are statistically significant at the 1% level.

The results above suggest that incumbents should realize that the probability of entry on a route has increased

Table 2
Probability of entry.
This table shows marginal effect estimates from probit regressions for entry of a low-cost carrier into a route in years t+1, t+2, and t+3 conditional on being entered at the second endpoint airport of the route (but not on the route itself) in year t. Possible routes are obtained from the T-100 Domestic Market database. Only years for which the low-cost carrier considered is actually operating are included. Year fixed effects are always included. Errors are clustered at the route level. t-statistics are reported in parentheses.

	Entry in year:			Obs.	Pseudo-$R2$
	t+1	t+2	t+3		
Southwest Airlines	0.2456***	0.061***	0.0538***	68,118	0.1496
	(35.20)	(9.86)	(9.45)		
JetBlue	0.1668***	0.0623***	0.083***	7,021	0.2518
	(8.28)	(3.35)	(3.10)		
Allegiant	0.1705***	0.0720***	0.0299***	28,881	0.0663
	(17.38)	(6.97)	(2.93)		
Frontier	0.2587***	0.0406***	−0.0177**	29,796	0.1145
	(22.80)	(4.76)	(−2.51)		
AirTran	0.1175***	0.0422***	0.0182**	28,161	0.1505
	(14.03)	(5.71)	(2.53)		
Virgin America	0.3345***	.	0.3171**	415	0.1896
	(3.08)	.	(2.02)		

when the low-cost carrier enters the second endpoint airport. However, entry into a single route would hardly disrupt the profitability of an incumbent. Therefore, I aggregate this route-level measure of threat at the airline level, looking at how many routes are threatened out of all those on which the incumbent operates. Importantly, I assign a different weight to different routes because routes with higher passenger traffic are more important for an airline given the higher number of paying passengers and their strategic nature (for instance, routes connecting to the hub in general have higher traffic, and incumbents are less likely to exit them). Data from the T-100 Domestic Market data set allow me to have information on the exact passenger traffic for each airline-route-year combination. Hence, I define the threat of entry in the following way:

$$Threat\ of\ entry_{i,t} = \sum_k \frac{Passengers_{k,i,t} \times I(Threatened\ route)_{k,i,t}}{Passengers_{i,t}}, \quad (1)$$

where $Passengers_{k,i,t}$ is the number of passengers for airline i, in year t, flying on route k, while $I(Threatened\ route)_{k,i,t}$ is an indicator function that takes a value of one if route k is under threat (because a low-cost airline just entered the second endpoint of the route in year t but has yet to enter the route itself) and takes a value of zero otherwise. Note that, as a result of constructing an airline-level measure of threat, the focus is going to be on the aggregate exposure of a carrier to entry threats by low-cost competitors and not on individual threats to single routes. The value of *Threat of entry* ranges from zero to one. A value of zero indicates that no routes for airline i in year t are under the threat of entry. A value of one indicates that all routes are under threat.

By measuring the percentage of routes where actual entry is realized at time t and regressing it on *Threat of entry* at time $t-1$, I find a coefficient of 14%, significant at the 1% level (this regression is unreported). This suggests that a one standard deviation increase in threat of entry (which is equal to 17 percentage points in my sample) is associated with a 2.5 percentage points increase in actual entry in the following year[10] (31% relative to the baseline ratio of 8.2%, see Table 1). To give an illustrative example, consider an incumbent that operates on 1,000 routes of equal passenger size for which threat of entry increases by one standard deviation. My estimates imply that low-cost airlines will enter on average 25 additional new routes immediately in the following year ($0.025 \times 1{,}000$). In other words, around 15% of the new threats (2.5/17) turn into additional entries in year t+1 (this number is roughly in line with the estimates computed directly from the route sample reported in Table 2, Column 1).

In my empirical analysis, I seek to estimate the effect of such a threat on the structure of debt maturity and leverage. Therefore, I run the following regression:

$$Debt\ maturity_{i,t} = \beta(Threat\ of\ entry_{i,t}) + \Gamma' X_{i,t} + \gamma_t + \gamma_i + \epsilon_{i,t}, \quad (2)$$

where $Threat\ of\ entry_{i,t}$ is defined as in Eq. (1) and captures the exposure of airline i in year t to the threat of entry posed by low-cost airlines. $X_{i,t}$ is a vector of time-varying controls. γ_t and γ_i are time and airline fixed effects. It is important that both time and airline dummies are included as my analysis focuses on the effect of cross-sectional variations in the threat of entry on debt maturity and I want to account for shocks affecting the entire industry in a similar way, as well as for time-invariant determinants. I run analogous regressions using leverage as the dependent variable.

[10] As $0.14 \times 0.17 \simeq 0.025$. The increase for the year thereafter is 1.3 percentage points.

4.2. Effects of entry

To understand why incumbents respond to entry threats, it is important to assess what the effect of the presence of low-cost carriers on profitability is. To do that, I exploit the Domestic Airline Consumer Airfare Reports issued by the U.S. Department of Transportation. Average fares are computed using a 10% random sample of all airline tickets issued by U.S. carriers, excluding charter air travel, covering 12 years of data.[11] Fares are based on the total ticket value, which consists of the price charged by the airlines plus any additional taxes and fees levied at the time of purchase.

Table 3 shows estimated coefficients for the logarithm of the average ticket price charged on each route regressed on a dummy variable that takes a value of one when a low-cost airline actually operates on the route and a value of zero otherwise. More precisely, the dependent variable is the average fare charged by air carriers operating on a given route recorded in the last quarter of the year and adjusted for inflation. In my regressions, I include time and route fixed effects to capture how ticket prices change *within* the same route when a low-cost airline is present. I cluster errors at the route level to account for time series correlation.

The presence of low-cost carriers has a dramatic effect on route profitability. Average fares are roughly 7% lower when a low-cost airline operates on a route. More specifically, the average fares charged on a given route are 4% lower when Southwest flies. The decrease in average ticket prices is 19%, 7%, 4%, and 12% for JetBlue, Allegiant, Frontier, and AirTran, respectively. In unreported calculations, I estimate that the expected cost at the airline level of a one standard deviation increase in the exposure to actual entry (estimated by regressing *ROA* on *Entry*, i.e., a measure that aggregates route-level threats at the airline level, see Appendix A) is roughly one percentage point: a 12.2% decrease in profitability relative to the baseline ratio of 8.2%.

The results reported above indicate that actual entry has a severe effect on the profitability of the incumbent, which in turn could limit its access to external funding. When entry coincides with the moment at which the incumbent needs to refinance a large fraction of maturing debt, a rationing of the credit supply may be associated with significant rollover failure costs. Besides the direct cost of restructuring the debt, related theoretical papers argue that rollover risk triggers additional real costs. For instance, Brunnermeier and Yogo (2009) and Diamond and He (2014) argue that rollover risk hinders real investments. While He and Xiong (2012) indicate that rollover risk increases credit risk. What are these costs in the case of airlines? A tangible example is provided by the decision of Moody's to downgrade LATAM Airlines in April 2016. To motivate the downgrade, the rating agency mainly refers to LATAM's exposure to rollover risk, emphasizing that "adverse market conditions are going to affect revenues and cash generation as the company faces meaningful near

[11] The data are published in the Bureau of Transportation Statistics' Passenger Origin and Destination (O&D) Survey, which covers years from 2000 to 2011.

Table 3
Low-cost airlines and route profitability.
The dependent variable in the regressions is the log of the average fare for each route. *Low − cost dummy* is a dummy variable that takes a value of one if at least one low-cost airline operates on the route and takes a value of zero otherwise. *Southwest dummy* is a dummy variable that takes a value of one if Southwest Airlines operates on the route and takes a value of zero otherwise. JetBlue, Allegiant, Frontier, AirTran, and Virgin America dummies have a similar interpretation. Average fares are obtained from the U.S. Department of Transportation Statistics' Passenger Origin and Destination (O&D) Survey. Fares are adjusted for inflation, cover the period from 2000 to 2011, and include a 10% sample of all airline tickets sold by U.S. carriers, excluding charter air travel. Average fares are average prices paid by all fare-paying passengers. Fares cover first class tickets paid to carriers offering such service but do not cover free tickets, such as those awarded by carriers offering frequent flyer programs. Time and route fixed effects are included and errors are clustered at the route level.

	Log ticket price							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Low-cost dummy	−0.0733***							
	(−11.12)							
Southwest dummy		−0.0447***						−0.0403***
		(−6.11)						(−5.51)
JetBlue dummy			−0.1981***					−0.1870***
			(−6.08)					(−5.55)
Allegiant dummy				−0.0650***				−0.0683***
				(−4.93)				(−5.15)
Frontier dummy					−0.0387***			−0.0398***
					(−4.44)			(−4.84)
AirTran dummy						−0.1174***		−0.1126***
						(−6.74)		(−6.45)
Virgin dummy							−0.1129**	0.0128
							(-2.23)	(0.17)
Route fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	75,525	75,525	75,525	75,525	75,525	75,525	75,525	75,525
R-squared	0.865	0.864	0.864	0.864	0.863	0.864	0.863	0.866

Table 4
Examples of real effects of rollover risk.
This table reports a number of selected examples of real effects of rollover risk. *Airline name* indicates the name of the incumbent airline, *Entry*(*%mkt*) indicates the percentage of routes—weighted by passenger traffic—where at least one new low-cost airline enters in year *t*, reported in Column 1. *Main entrant* indicates which low-cost airline enters the largest fraction of the incumbent airline's market in year *t*. *Rollover* indicates the percentage of long-term debt that the incumbent needs to refinance during year *t* (as reported at the beginning of the year). Δ*issuance*, Δ*CAPX*, Δ*fleet*, Δ*emp*., and Δexp. indicate, respectively, the annual percentage change with respect to the previous year in dollar long-term debt issuance, capital expenditures, fleet size, number of employees, and expenditures per available seat-mile (computed, for instance, as $(CAPX_t/CAPX_{t-1} - 1) \times 100$). *NA* indicates that the value is not available. Data are obtained from the T-100 Domestic Market database and Compustat.

	Airline information				Effects (% change from previous year)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Airline name	Year	Main entrant	Entry(%mkt)	Rollover	ΔIssuance	ΔCAPX	ΔFleet	ΔEmp.	ΔExp.
1) Northwest Airlines	2005	AirTran	0.13	0.09	−70	−25	−40	−19	NA
2) ExpressJet	2008	Frontier	0.15	0.94	NA	−79	−11	−39	−14
3) Republic Airlines	2012	AirTran	0.11	0.12	−90	−61	0	−3	33
4) Vanguard Airlines	2001	Frontier	0.12	0.35	−56	−24	0	−3	−6
5) SkyWest Airlines	2012	AirTran	0.16	0.11	−41	−65	2	−1	−9

term debt maturities and capital commitments." To ease liquidity concerns, LATAM Airlines has announced: i) a reduction of its passenger capacity, ii) a consistent workforce and cost reduction, and iii) a cut in planned capital expenditures (which mostly consist of fleet commitments).[12]

By avoiding to refinance under critical conditions, firms may prevent inefficient corporate downsizing. In my sample, there are few refinancing episodes that coincide with large entry by low-cost carriers (consistent with the hypothesis that incumbents do extend debt maturity before actual entry occurs). I report a number of those in Table 4 as anecdotal evidence. Specifically, the table shows annual percentage changes in key corporate quantities for a number of incumbent airlines facing new low-cost competition while, at the same time, having to refinance large amounts of debt (the fraction of long-term debt that needs to be rolled over during the year is reported in Column 4). In all cases, dollar debt issuance decreases with respect to the previous year (see Column 5), consistent with the hypothesis that debt becomes relatively more expensive or is rationed. This, in turn, seems to have important (even though heterogeneous) consequences for corporate policies. Similar to the LATAM Airlines example, most airlines significantly cut capital expenditures and reduce the size of the workforce (see Columns 6 and 8).[13] In two out of

[12] See "Moody's downgrades LATAM Airlines CFR to B1 rating; outlook stable," Moody's rating action, disclosed on April 29, 2016.

[13] The case of Northwest Airlines seems particularly relevant. Fitch downgraded the airline in June 2005, as mounting competition by low-cost carriers and increasing oil price made it unlikely for Northwest to be

five cases, the incumbents also reduce the size of their fleet (see Column 7), suggesting that the airlines could be forced to sell assets. In three other cases (see Column 9), the air carriers reduce the expenditures per available seat-mile (which could indicate a reduction of in-flight services such as magazines, newspapers, snacks, Wi-Fi, etc.). While this evidence does not provide a conclusive link between rollover risk and real effects, it is nonetheless suggestive of high rollover failure costs. These costs, in turn, provide a rationale for extending debt maturity before actual entry occurs. A more systematic investigation supporting the arguments put forward in this section is provided in the Online Appendix.

5. Empirical results

This section presents the main empirical results of the paper. Section 5.1 explores the effect of the threat of entry on debt maturity, providing evidence in support of the rollover risk channel. Section 5.2 looks at the effect on leverage, Section 5.3 considers airlines' heterogeneity, and Section 5.4 provides robustness tests and rules out alternative explanations.

5.1. Threat of entry and debt maturity

In this section, I test the hypothesis that the threat of entry by low-cost competitors has an effect on debt maturity (Hypothesis I). The rationale for adapting debt maturity ex ante is provided by the disruptive effects of actual entry (the average profitability decreases by one percentage point, 12.2% relative to the baseline ratio of 8.2%, see Section 4.2). This hypothesis also follows from the theoretical framework developed by Brunnermeier and Yogo (2009). In my empirical specification, I include airline fixed effects to account for time-invariant firm characteristics and I exploit that the threat of entry is computed at the firm (i.e., not at the industry) level creating cross-sectional heterogeneity, as different airlines are exposed to entry threats to different degrees. Errors are clustered at the firm level because corporate variables are in general persistent and observations are likely to be correlated in the time series. Table 5 reports the coefficients I obtain from the regressions.

The estimated effect of *Threat of entry* on debt maturity is positive and statistically significant at the 1% level (t-statistic of 3.23). However, *Threat of entry* could be correlated with known predictors of debt maturity such as size, asset maturity, tangibility, and profitability (see, e.g., Guedes and Opler, 1996). Additionally, it could be affected by a change in investment opportunities or a shift in the demand for transportation. Hence, in the specifications reported in Columns 2 and 3, I include several time-varying

Table 5
Threat of entry and debt maturity.
This table presents results from regressions of *Debt maturity* on *Threat of entry* and controls. *Threat of entry* measures the percentage of routes under threat weighted by the number of paying passengers. *Debt maturity* is defined as the fraction of long-term debt maturing in more than three years following Barclay and Smith (1995) and Custódio et al. (2013). The sample consists of all passenger airlines in Compustat that could be matched to the T-100 Domestic Market database. All other variables are described in Table A.1 in Appendix A. All regressions include an intercept (not reported) and year and airline dummies when indicated. Standard errors are clustered at the airline level. t-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Debt maturity			
Model:	(1)	(2)	(3)	(4)
Threat of entry	0.4158***	0.3855***	0.3562***	0.2445**
	(3.23)	(3.78)	(4.44)	(2.66)
Sales		0.1806***	0.1383*	0.1312**
		(3.36)	(1.95)	(2.21)
Asset maturity		0.0139*	0.0106	0.0118*
		(1.99)	(1.65)	(2.05)
Tangibility			0.1950	0.1344
			(0.93)	(0.64)
Profitability			0.1517	0.0792
			(0.59)	(0.26)
ΔPassengers			−0.0667	−0.0765
			(−1.29)	(−1.25)
Chapter 11				−0.0569
				(−1.30)
Q				0.0043
				(0.07)
Trend				−0.0149**
				(−2.15)
Credit spread				0.0744
				(1.39)
Term spread				−0.0106
				(−0.94)
Airline fixed effects	Y	Y	Y	Y
Time fixed effects	Y	Y	Y	N
Observations	239	239	239	239
R-squared	0.536	0.608	0.621	0.582

controls. Among the control variables, *Log sales* and *Asset maturity* are always significantly correlated with *Debt maturity*, in line with results in Barclay and Smith (1995) and Benmelech (2009). Furthermore, I find positive but statistically insignificant coefficients for *Tangibility* and *Profitability*. Column 4 additionally includes *Chapter* 11, a dummy variable indicating when an airline is in financial distress; Tobin's q (Q), as a proxy for investment opportunities; and variables capturing the trend in debt maturity and the credit and term spreads (the time fixed effects are dropped in this last specification as they would be collinear with the last three variables considered).

Overall, the threat of entry has a positive and statistically significant effect on debt maturity in all specifications. A one standard deviation increase in *Threat of entry* is associated with an increase of 6.0 percentage points in the proportion of long-term debt held by incumbent airlines (which corresponds to an increase of 10% from the baseline of 60%).[14] Fig. 4 illustrates the average debt

able to roll over more than 30% of the debt maturing in the following two years. This, in turn, had a number of important implications. For example, Northwest's access to debt markets without posting collateral was precluded. Furthermore, negotiations to cut labor costs were initiated, and the airline started a broader program to reduce total costs (see "Fitch downgrades Northwest Airlines to D," *BusinessWire* – September 15, 2015 and "NWA debt downgraded; Union says airline hired replacements," *USA Today* – June 2, 2005).

[14] The economic magnitude reported is obtained using specification 3 of Table 5 (i.e., the specification that includes both airline and time fixed

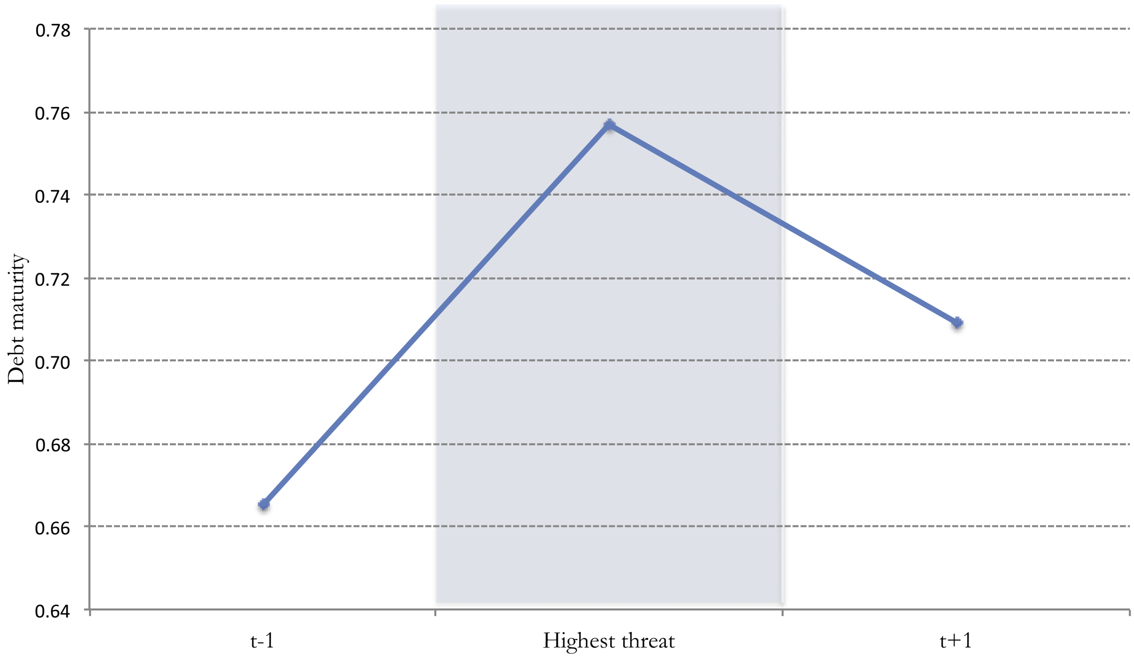

Fig. 4. Average debt maturity around High threat episodes. This figure illustrates average debt maturity levels in the years around the highest threat exposure for the airlines in my sample. I consider only airlines that i) are not in distress (i.e., *Chapter* 11 = 0) and ii) for which the highest threat level is at least one standard deviation above the mean of the sample. *Debt maturity* is defined as the fraction of long-term debt maturing in more than three years following Barclay and Smith (1995) and Custódio et al. (2013). Data are obtained from the T-100 Domestic Market database and Compustat. The sample period goes from 1991 to 2014.

maturity levels around the year of biggest threat for the airlines in my sample: incumbents increase debt maturity by almost ten percentage points on average, roughly in line with the estimates from the multivariate analysis. This implies that, by responding to entry threats, airlines will have to refinance a lower fraction of debt in the following three years. This result is consistent with the prediction that firms issue bonds of the shortest maturity as long as subsequent rollover is guaranteed and bonds of longer maturity when facing rollover risk (see Hypothesis I).[15]

Furthermore, I explore the effect of the threat of entry on alternative proxies of debt maturity. Results reported in Table 6 provide estimates for the effect on newly issued long-term debt (*Long − term debt issuance*) and on the proportion of long-term debt maturing in more than five years (*Maturity*5). All coefficients are positive and statistically different from zero. Airlines in my sample issue on average 7.8% of new long-term debt per year, while the portion of long-term debt that reaches the maturity date is 4.5% and the fraction that is bought back is 6% (all numbers are scaled by book assets). As a result, the fraction of long-term debt outstanding is, on average, slowly decreasing over time; this finding is consistent with Custódio et al. (2013) and Harford et al. (2014). Coefficients in Column 3 indicate that an increase of one standard deviation in the threat of entry is associated with an average increase in long-term debt issuance of 3.0 percentage points when both airline and time fixed effects are included (the net effect on leverage is discussed in the next section).

When looking at different measures of maturity, the estimated coefficient for the effect of *Threat of entry* on *Maturity* is 0.24 (see Table 5, Column 4) and the effect on *Maturity*5 is 0.28 (see Table 6, Column 5). This implies that a one standard deviation increase in *Threat of entry* is associated with increases of, respectively, 4.1 and 4.8 percentage points[16] for *Maturity* and *Maturity*5. This indicates that threatened airlines mostly issue debt of maturity longer than five years. Consistently, the estimated coefficient for the effect of the threat of entry on the portion of debt maturing exactly in four or five years is negative and non-statistically different from zero (this result is unreported). Overall, results in this section are suggestive of an active management of rollover risk in the presence of entry threats. Alternative explanations are considered in Section 5.4.

effects). This analysis uses the smaller set of 239 observations for which all control variables are available (see the discussion in Section 3.1). The effect estimated for an analogous specification obtained from the broadest set of observations available is around 4.5 percentage points, almost 25% lower. The selection effects are further discussed in the Online Appendix.

[15] Xu (2018) shows that speculative-grade firms often refinance early to hedge against refinancing risk. This finding provides support for the argument that risky firms actively manage debt maturity to avoid rolling over their debt in bad times. Data in my sample indicate that the average speculative-grade airline needs to refinance 43% of its long-term debt within the next three years, and 16% within one year. This suggests that, even though some airlines may refinance early, it remains crucial for them to further extend debt maturity in the event of entry threats.

[16] The increase for *Maturity*5 is 12% from the baseline of 41%.

Table 6
Alternative measures of debt maturity.
This table presents results from regressions of *Long – term debt issuance* and *Maturity5* on *Threat of entry* and controls. *Threat of entry* measures the percentage of routes under threat weighted by the number of paying passengers. The sample consists of all passenger airlines in Compustat that could be matched to the T-100 Domestic Market database. All other variables are described in Table A.1 in Appendix A. All regressions include an intercept (not reported) and year and airline dummies when indicated. Standard errors are clustered at the airline level. *t*-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Long-term debt issuance				Maturity5
Model:	(1)	(2)	(3)	(4)	(5)
Threat of entry	.1824**	.1832***	.1779***	.1107*	.2809**
	(2.68)	(2.84)	(3.23)	(1.77)	(2.43)
Sales		−0.0232	−0.0202	−0.0259	.1771***
		(−0.50)	(−0.35)	(−0.47)	(2.87)
Asset maturity		−0.0005	−0.0018	−0.0022	.0170**
		(−0.13)	(−0.35)	(−0.41)	(2.60)
Tangibility			0.0445	0.0807	0.1194
			(0.31)	(0.63)	(0.59)
Profitability			−0.0182	−0.0390	0.1883
			(−0.17)	(−0.36)	(0.89)
ΔPassengers			0.0083	−0.0034	−0.0314
			(0.34)	(−0.14)	(−0.54)
Chapter 11				−0.0046	−0.0565
				(−0.21)	(−0.91)
Q				0.0001	−0.0892
				(0.00)	(−1.35)
Trend				0.0013	−.0262***
				(0.28)	(−3.69)
Credit spread				0.0187	0.0262
				(0.77)	(0.91)
Term spread				−0.0019	−0.0112
				(−0.28)	(−0.90)
Airline fixed effects	Y	Y	Y	Y	Y
Time fixed effects	Y	Y	Y	N	N
Observations	239	239	239	239	239
R-squared	0.274	0.279	0.281	0.191	0.648

Table 7
Threat of entry and leverage.
This table presents results from regressions of (book) *Leverage* on *Threat of entry* and controls. *Threat of entry* measures the percentage of routes under threat weighted by the number of paying passengers. *Leverage* is defined as total debt over total book assets. The sample consists of all passenger airlines in Compustat that could be matched to the T-100 Domestic Market database. All other variables are described in Table A.1 in Appendix A. All regressions include an intercept (not reported) and year and airline dummies when indicated. Standard errors are clustered at the airline level. *t*-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Leverage			
Model:	(1)	(2)	(3)	(4)
Threat of entry	0.0849	0.0643	0.0022	0.0153
	(0.59)	(0.53)	(0.02)	(0.14)
Sales		0.0469	0.0240	0.0080
		(0.72)	(0.35)	(0.12)
Asset maturity		.0086*	−0.0024	−0.0034
		(1.89)	(−0.40)	(−0.57)
Tangibility			.4805**	.5253**
			(2.60)	(2.74)
Profitability			0.0670	0.0178
			(0.33)	(0.09)
ΔPassengers			0.0009	−0.0199
			(0.03)	(−0.73)
Chapter 11				−0.0709
				(−0.73)
Q				0.0470
				(0.93)
Airline fixed effects	Y	Y	Y	Y
Time fixed effects	Y	Y	Y	Y
Observations	239	239	239	239
R-squared	0.639	0.665	0.703	0.711

5.2. Threat of entry and leverage

Table 7 reports results for regressions of book leverage on *Threat of entry* and controls. I consider book leverage (instead of market leverage) because the increase in the probability of entry could already be discounted into the stock price of the incumbent airline thereby inducing a mechanical effect on market leverage. Additionally, Graham and Harvey (2001) indicate that chief financial officers (CFOs) mostly decide debt-equity ratios based on book values as market values fluctuate daily and are difficult to target.

The results suggest that the threat of entry does not have a significant effect on book leverage. This evidence is consistent with previous findings in the literature suggesting that the current capital structure is strongly related to historical market values (Baker and Wurgler, 2002), leverage is highly persistent (e.g., Lemmon et al., 2008), and mostly firm-specific (MacKay and Phillips, 2005). This result is also in line with Graham and Harvey (2001), as in a survey of 392 financial executives the authors find little evidence that product market factors affect debt to equity ratios. Importantly, I do not claim that incumbent airlines would not benefit from having lower leverage when competition increases but merely that to adjust capital structure in the short-run may be difficult.

Interestingly, most of the time-varying controls I consider have at best a marginal effect on book leverage when both time and airline fixed effects are included. Therefore, I conclude that, in my sample, most of the variation in book leverage appears to be driven by airlines' time-invariant specificities. Overall, the results provided in this section suggest that the threat of entry does not have a causal effect on leverage, thereby rejecting Hypothesis II. I cannot however rule out the possibility that the limited sample size decreases the power of my tests. To mitigate this concern, in the Online Appendix, I replicate my analysis on a larger sample of 755 observations from public and private airlines, using coarser financial variables. The results are qualitatively similar.

5.3. Heterogeneous maturity extension and future investments

Goolsbee and Syverson (2008) show that the major airlines cut ticket prices in response to route-level threats, either to deter or accommodate entry. However, smaller airlines do not always possess the necessary financial resources to respond to entry threats by implementing aggressive pricing policies. This implies that a financially constrained airline should find it relatively easier, and more convenient, to adapt its maturity structure rather than to commit to keep prices low (as such a commitment would hardly be credible when the new entrant has "deeper pockets"). From a theoretical perspective, the effect of threat of entry on debt maturity should therefore be driven by those firms that are particularly exposed to rollover risk and lack the necessary financial resources to engage in price wars (see Diamond, 1991a).

In the following, I focus on the response to entry threats of firms that are financially constrained or whose long-term debt is rated as speculative. The intuition is that these firms are more exposed to the risk of default and would experience more difficulties in finding alternative sources of financing in the event of a rollover failure (see Hypothesis III). In support of this categorization, Harford et al. (2014) show empirically that financially constrained firms are more likely to implement a conservative financial strategy that accounts for the possibility of a rollover failure and Diamond (1991a) derives theoretically the prediction that badly rated firms should issue at a longer maturity to reduce rollover risk. In addition, Xu (2018) shows that, in good times, speculative-grade firms frequently refinance early in order to hedge against refinancing risk. Consistent with the arguments above, rating agencies have recently started to consider rollover risk exposure in their determination of total risk (see He and Xiong, 2012, p. 415).

I define financially constrained firms using the SA index from Hadlock and Pierce (2010). The SA index is a function of a firm's size and age, the authors show that such a measure is better suited for identifying financially constrained firms than alternative measures such as the Kaplan-Zingales index (KZ index). Firms in the top decile of the SA index are defined as financially constrained. I define high credit quality airlines (long-term debt rated BBB- or higher) using rating data obtained from Standard and Poor's. According to the arguments discussed above the response to entry threats should be weaker for investment-grade airlines and stronger for financially constrained incumbents. Consistent with these predictions, results reported in Table 8 indicate that investment-grade airlines basically do not alter debt maturity in response to entry threats. By contrast, conditional on an airline being financially constrained, the marginal effect of a one standard deviation increase of threat of entry on debt maturity is 16 percentage points (according to the coefficients reported in Column 3) versus six percentage points for the average airline (while, on average, *Debt maturity* is nine percentage points lower for financially constrained airlines).[17] Overall, results from the heterogeneity tests corroborate the rollover risk hypothesis.

Furthermore, I explore the effect of a change in debt maturity on future investment when the low-cost airlines are threatening entry in a large portion of the incumbent airline's market.[18] In my analysis, I estimate the effect of a change in debt maturity at time t on investment at time t+1, conditional on an airline being exposed to the highest decile of threat measured at time t. I use the broadest set of available observations to have enough statistical power.

Table 8
Heterogeneous response to entry threats.
This table presents results from regressions of *Debt maturity* on *Threat of entry* and interactions of *Threat of entry* and proxies for the exposure to rollover risk. *Threat of entry* measures the percentage of routes under threat weighted by the number of paying passengers. *Debt maturity* is defined as the fraction of long-term debt maturing in more than three years following Barclay and Smith (1995) and Custódio et al. (2013). *Investment grade* is a variable that takes a value of one if the rating of the long-term debt for an airline is BBB- or higher. *High SA* is a dummy variable that identifies the firms in the top decile of the *SA* index distribution (i.e., the 10% most financially constrained observations). The *SA* index is computed following Hadlock and Pierce (2010) as $-$ 0.737 $\times$ Size + 0.043 $\times$ Size^2 $-$ 0.040 $\times$ Age. The sample consists of all passenger airlines in Compustat that could be matched to the T-100 Domestic Market database. All other variables are described in Table A.1 in Appendix A. All regressions include an intercept (not reported) and year and airline dummies when indicated. Standard errors are clustered at the airline level. t-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Debt maturity		
Model:	(1)	(2)	(3)
Threat of entry × Investment grade	−.4935***		−.4623***
	(−4.67)		(−4.39)
Investment grade	0.2125***		0.2109***
	(3.52)		(3.62)
Threat of entry × High SA		.6527**	.6027**
		(2.40)	(2.12)
High SA		−0.0991	−0.0925
		(−1.67)	(−1.60)
Threat of entry	.3870***	.3215***	.3520***
	(3.90)	(3.57)	(3.31)
Sales	.1294*	.1327*	.1234*
	(1.98)	(1.92)	(1.97)
Asset maturity	0.0099	.0115*	.0108*
	(1.56)	(1.91)	(1.80)
Tangibility	0.1950	0.1531	0.1551
	(0.96)	(0.75)	(0.78)
Profitability	0.1628	0.1351	0.1492
	(0.65)	(0.53)	(0.61)
ΔPassengers	−0.0730	−0.0767	−0.0820
	(−1.40)	(−1.60)	(−1.67)
Airline fixed effects	Y	Y	Y
Time fixed effects	Y	Y	Y
Observations	239	239	239
R-squared	0.635	0.627	0.640

Results reported in Table 9 indicate that incumbents invest, on average, 3.9 percentage points less in the year following a severe threat. This negative effect can be mitigated by extending debt maturity. When a threatened airline increases Δ*Maturity* by one standard deviation in year t, the reduction in investment at time t+1 is more than halved. Specifically, an increase by 19 percentage points in Δ*Maturity* (which corresponds to a one standard deviation increase in Δ*Maturity* in my sample) leads to an increase of 2.4 percentage points in future investment (the net effect on investment being 2.4– 3.9 = −1.5 percentage points versus −3.9 percentage points when maturity is not adjusted). This finding is consistent with previous research. Almeida et al. (2012) show that firms with large portions of debt maturing right at the time of the crisis had to cut investment more severely. This suggests that the same firms would have been better off by lengthening debt maturity just before the financial crisis hit, in line with the arguments put forward in this paper.

From a cost-benefit perspective, I estimate that an increase of 19 percentage points in debt maturity is

[17] This last estimation is however non-statistically significant, presumably due to the limited sample size.

[18] As, according to a growing theoretical literature, rollover risk can lead to underinvestment see, e.g., Brunnermeier and Yogo (2009) and Diamond and He (2014).

Table 9
Effects on future investment.
This table presents results from regressions of *Investment* in year t+1 on *High threat*, Δ*Maturity*, and controls at time t. *High threat* identifies the incumbents exposed to the highest decile of *Threat of entry*. *Threat of entry* measures the percentage of routes under threat weighted by the number of paying passengers. *Debt maturity* is defined as the fraction of long-term debt maturing in more than three years following Barclay and Smith (1995) and Custódio et al. (2013). Δ*Maturity* is the difference between *Debt maturity* at time t and *Debt maturity* at time t−1. The sample consists of all passenger airlines in Compustat that could be matched to the T-100 Domestic Market database. All other variables are described in Table A.1 in Appendix A. All regressions include an intercept (not reported) and year and airline dummies when indicated. Standard errors are clustered at the airline level. t-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Investment(t+1)		
Model:	(1)	(2)	(3)
High threat × ΔMaturity	.1394**	.1462**	.1224**
	(2.64)	(2.67)	(2.00)
High threat	−.0385**	−.0389**	−.0390*
	(−2.08)	(−2.12)	(−1.87)
ΔMaturity	−0.0260	−0.0255	−0.0189
	(−0.85)	(−0.84)	(−0.59)
Sales		0.0045	0.0011
		(0.13)	(0.03)
Asset maturity		0.0009	0.0012
		(0.31)	(0.26)
Tangibility			−0.0024
			(−0.02)
Profitability			0.0937
			(0.88)
ΔPassengers			0.0111
			(0.47)
Airline fixed effects	Y	Y	Y
Time fixed effects	Y	Y	Y
Observations	249	249	249
R-squared	0.516	0.517	0.525

associated with 22 basis points higher interest expenses (scaled by book assets), which is an 8% increase with respect to the average cost of debt.[19] Besides higher interest expenses, the choice of issuing long-term debt also entails some non-monetary costs. For instance, by issuing long-term debt the borrower is forced to commit to a specific debt maturity, which makes it more difficult to quickly readjust its liability structure (Brunnermeier and Yogo, 2009). Furthermore, a longer debt maturity increases agency costs (Myers, 1977) and sends a negative signal to the lenders (Flannery, 1986). These costs are, however, arguably moderate if compared to an additional reduction in capital expenditures of 2.4 percentage points when maturity is not adjusted (see above), thereby providing a rationale for extending debt maturity ex ante. Overall, results in this section suggest that threatened airlines find it optimal to trade off the benefits of short-term financing for lower rollover failure costs.

5.4. Robustness

5.4.1. Mechanical adjustments

A relevant question to address is whether these results can arise mechanically. Chen and Zhao (2007) and Chang and Dasgupta (2009) demonstrate that the leverage ratio of a firm tends to revert to the mean mechanically regardless of the firm's financing preference. This implies that, unless a firm adopts an extreme financing policy, a high leverage ratio tends to decrease, while a low leverage ratio tends to increase. Similar dynamics may be at work with debt maturity in the setting of this paper. For instance, if incumbents borrowing at short-maturities in year t-1 were more likely to be threatened by low-cost carriers, a positive correlation between the variables *Threat of entry* and *Debt maturity* could arise regardless of their maturity preference. Importantly, this is unlikely to be the case, as I find no correlation between threat of entry and past debt maturity levels in my sample.

In order to provide further evidence against the presence of a mechanical relation, I follow the approach employed by Chang and Dasgupta (2009) and Danis et al. (2014). Specifically, I explore whether a correlation between the variables *Threat of entry* and *Debt maturity* emerges in simulated samples in which airlines choose debt maturity randomly. The idea in a nutshell is that if I were to find similar results in the actual and simulated samples, then the correlation between maturity and entry threats would likely be mechanical.

Similar to Chang and Dasgupta (2009) and Danis et al. (2014), I perform the following procedure. First, I compute actual financing deficits—defined as the difference between the change in assets and the change in retained earnings—for all airline observations in my sample. Second, I take initial maturities as in the original sample and I construct a series of maturity levels (always bounded to be between zero and one) that evolve according to the following rule: each firm issues debt of amount equal to the financing deficit (or buys it back in case the financing deficit is negative) and *random* maturity. The maturity at the end of each period is, therefore, the combination of the maturity at the beginning of the period and the maturity of the new debt issue. In other words, in the simulated samples, maturity evolves following a random path in which the size of the financing transaction is determined by the actual financing deficits but the choice between short-term and long-term debt is random (each having a 50% chance).

Eventually, I obtain for each airline a time-series of simulated maturity levels. I keep all other variables as in the original sample. Importantly, I run two simulation procedures. In the first, the firm always issues or buys back debt (i.e., I assume that equity is never issued or repurchased) and the only random decision is the one about maturity.[20] In the second simulation, the firm decides first between

[19] I measure interest expenses as "interest and related expenses" (*xint*) over total assets (*at*). I scale interest expenses by assets to make the magnitude comparable to investment (which is also scaled by assets). The number reported above is obtained by regressing the change in interest expenses on the lagged change in debt maturity (this regression is unreported). Importantly, this analysis is only suggestive as it implicitly assumes that airlines that do and do not extend debt maturity before entry are similar along all other dimensions. Threatened airlines that do not extend debt maturity are, however, likely to face even higher issuance costs ex ante (which, in turn, lead them not to extend debt maturity before entry occurs).

[20] The preference for adjusting debt over equity is consistent with the pecking order and with the empirical evidence (see Chang and Dasgupta, 2009).

Table 10
Mechanical adjustments.
Simulated data used to estimate the coefficients reported in Column 1 are obtained assuming that firms issue (buy back) short-term debt of size equal to the *actual* financing deficit (surplus) with probability 50%, and long-term debt with probability 50%. Simulated data used to estimate the coefficients reported in Column 2 are obtained assuming that: first, firms decide randomly between equity and debt financing (each having a probability of 50%). Second, in the case debt is selected, firms decide between short- and long-term debt randomly (each having a probability of 50%). The simulation procedure is explained in detail in Section 5.4.1. For simulated samples, the reported parameter estimates represent the average coefficients obtained from 500 replications of the simulation, 95% confidence intervals are included in square brackets.

	Debt maturity	
	(1) S($p_{debt}=1$, $p_{short}=0.5$)	(2) S($p_{debt}=0.5$, $p_{short}=0.5$)
Threat of entry	0.0281	0.0095
	[−0.36, 0.42]	[−0.33, 0.36]
Sales	−0.0394	−0.0544
	[−0.23, 0.15]	[−0.27, 0.15]
Asset maturity	0.0088	0.0069
	[−0.01, 0.03]	[−0.02, 0.03]
Tangibility	−0.1631	0.0775
	[−0.83, 0.52]	[−0.70, 0.72]
Profitability	0.2341	0.2994
	[−0.25, 0.66]	[−0.21, 0.69]
ΔPassengers	−0.0347	−0.0449
	[−0.12, 0.05]	[−0.14,0.05]
Airline fixed effects	Y	Y
Time fixed effects	Y	Y
Observations	239	239

debt and equity financing. If equity is issued (with probability 50%), debt maturity does not change from the previous period. If the firm decides to issue debt (in the remaining 50% of the cases), than the firm tosses another coin to choose between long-term and short-term debt.

Coefficients obtained regressing the variable debt maturity on threat of entry in the actual and simulated samples are reported in Table 10. Column 1 reports the average coefficients obtained from 500 replications of the procedure when firms never issue or repurchase equity, Column 2 when firms choose randomly between equity and debt. The estimated coefficients for threat of entry in the simulated samples are very different from those estimated on the actual data (see the coefficients reported in Table 5 as a comparison). More importantly, the 95% confidence intervals always include the zero, indicating that the positive relation between maturity and entry threats does not arise when debt maturity decisions are random. Overall, results in this section suggest that the correlation between the variables *Debt maturity* and *Threat of entry* is unlikely to arise mechanically.

5.4.2. Actual entry versus threat of entry

A potential concern with the identification arises because threat of entry and actual entry can be highly correlated once the route-level measures are aggregated at the airline level. In this section, I provide evidence against the possibility that the variable *Threat of entry* simply proxies for actual entry. To do that, I run a horse race between the variables *Threat of entry* and *Entry*. In particular, I construct *Entry* as follows:

$$Entry_{i,t} = \sum_k \frac{Passengers_{k,i,t} \times I(Route\,entry)_{k,i,t}}{Passengers_{i,t}}, \tag{3}$$

where $Passengers_{k,i,t}$ is the number of passengers for airline i, in year t, flying on route k, while $I(Route\,entry)_{k,i,t}$ is an indicator function that takes a value of one if a new low-cost airline enters exactly on route k in year t and takes a value of zero otherwise. The results are reported in Table 11, Column 1. The evidence from the table clearly indicates that the effect of *Threat of entry* on debt maturity remains strongly significant when controlling for actual entry (the estimated coefficient is 0.31 versus 0.36 in the baseline specification). Overall, the magnitudes of the coefficients of *Entry* and *Threat of entry* are very similar, thereby suggesting that the effect of expected competition on debt maturity is as large as the effect of actual competition.

Furthermore, I also replicate my analysis controlling for *Recent entry*, i.e., a more persistent version of the entry variable:

$$Recent\ entry_{i,t} = \sum_k \frac{Passengers_{k,i,t} \times I(Recent\ route\,entry)_{k,i,t}}{Passengers_{i,t}}, \tag{4}$$

where $I(Recent\,route\,entry)$ is defined as a dummy variable that takes a value of one if entry has occurred in any of the two previous years (i.e., at any time during year t or $t-1$). The coefficient for *Threat of entry* barely changes (see Table 11, Column 2). More persistent variations of *Recent entry* also leave the coefficient of *Threat of entry* largely unchanged (these regressions are unreported). Overall, results in this section confirm that entry threats are directly related to debt maturity.

5.4.3. Crises

An important question that remains unanswered is whether my results hold only during periods of intense market stress. To test whether this is the case, I reproduce the main analysis excluding observations from recession years, as identified by the National Bureau of Economic Research (NBER)'s Business Cycle Dating Committee. This reduces by 16% the size of my sample. The coefficient for the effect of *Threat of entry* on *Maturity* estimated in a sample excluding crises is reported in Table 11, Column 3. The resulting estimate is substantially unchanged: 0.33 versus 0.36 in the baseline specification (see Table 5, Column 3), still significant at the 1% level. This rules out the possibility that the results might be driven by crisis periods.

5.4.4. Price dynamics

Goolsbee and Syverson (2008) and Gedge et al. (2014) suggest that incumbent airlines lower ticket prices when their routes are threatened to either accommodate or deter entry. In this section, I provide evidence that rules out the possibility that debt maturity changes are driven by price adjustments in anticipation of actual entry. To account for price adjustments, I replicate my main analysis including proxies for the average ticket price (*Avg. ticket price* estimated as *Passenger revenues* scaled by *#Paying Passengers*) and the change in the average ticket price

Table 11
Robustness.
This table presents results from regressions of *Debt maturity* on *Threat of entry* (or *Acela threat*) and controls. *Entry* measures the percentage of the incumbent airline's routes, weighted by the number of paying passengers, where a new low-cost airline is entering; *Recent entry* measures the percentage of routes in which a new low-cost airline has just entered or has entered in the previous year. The specification reported in Column 3 excludes NBER recession years. *Avg. ticket price* is passenger revenues over revenue passengers carried; Δ*ticket price* is the change in *Avg. ticket price* from the previous year. $E[\Delta passengers]$ is the percentage annual change in passenger traffic from year t to year $t+1$. *Threat* (*excl. rural*) measures the percentage of routes under the threat of entry excluding routes connecting rural airports. Rural airports are airports that have fewer than 100,000 commercial passengers departing from the airport by air and satisfy at least one of the following conditions: the airport is not located within 75 miles of another airport from which 100,000 or more commercial passengers departed, the airport was receiving essential air service subsidies as of August 5, 1997, or the airport is not connected by paved roads to another airport. *Acela threat* is a dummy variable that takes a value of one from 1999 onwards for airlines highly exposed (above 90% percentile) to Acela Express routes. All other variables are described in Table A.1 in the Appendix. All regressions include an intercept (not reported) and year and airline dummies when indicated. Standard errors are clustered at the airline level. *t*-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Entry (1)	Recent entry (2)	Excl. crises (3)	Price adj. (4)	E[Demand] (5)	Rural routes (6)	Acela (7)
Threat of entry	.3097***	.3144***	.3326***	.4768***	.3465***		
	(3.23)	(3.44)	(3.40)	(4.02)	(4.38)		
Threat (excl. rural)						.3538***	
						(4.37)	
Acela threat							.2086**
							(2.82)
Entry	0.3550						
	(1.51)						
Recent entry		.3675**					
		(2.04)					
Avg. ticket price				0.0004			
				(0.38)			
ΔTicket				−0.0285			
				(−0.08)			
E[ΔPassengers]					0.0561		
					(0.59)		
Controls	Y	Y	Y	Y	Y	Y	Y
Airline fixed effects	Y	Y	Y	Y	Y	Y	Y
Time fixed effects	Y	Y	Y	Y	Y	Y	Y
Observations	239	239	203	146	228	239	238
R-squared	0.628	0.634	0.625	0.638	0.631	0.621	0.536

charged by the incumbent (Δ*Ticket price*). If the change in debt maturity merely reflects an adjustment in fares, I should find that the estimated effect of the threat of entry on debt maturity weakens significantly, while average ticket prices and/or changes in ticket prices should be strongly correlated with debt maturity. However, the coefficient estimated for threat of entry remains economically and statistically significant when I account for variations in fares, while the coefficients estimated for Δ*Ticket price* and *Avg. ticket price* are not statistically different from zero (see Table 11, Column 4). The number of observations for this test is lower than that in the baseline analysis because the variable *Passenger revenues* is not always available. If I replicate the main analysis on this smaller sample—without controlling for price variations—I get a coefficient of 0.48 for the effect of threat of entry on debt maturity (analogous to the specification that includes price controls). Therefore, I conclude that changes in debt maturity are unrelated to *contemporaneous* ticket price adjustments. However, this does not exclude that an increase in debt maturity could allow the incumbent to become more flexible in setting lower prices in the future (see the Online Appendix).

5.4.5. Demand channel

An additional concern with the findings is that the low-cost airline could enter the second end-point airport of a route anticipating an increase in the demand of transportation on the route itself. I therefore include in the main specification the change in passenger traffic from year t to year $t+1$ as a control for the expected change in passenger traffic (under the assumption that the low-cost airline can predict future passenger traffic with high precision). Results are reported in Table 11, Column 5 and are unchanged (the estimated coefficient is 0.35 versus 0.36 in the baseline specification). Additional evidence against a demand channel is provided in the Online Appendix.

5.4.6. Rural airports

Another question that arises naturally is whether low-cost airlines pose a disproportionately high threat to routes connecting small or rural airports. To rule out this possibility, I provide evidence against the possibility that the documented effects are driven by the geographical distribution of the incumbent airlines. To do that, I use the list of rural airports published by the Bureau of Transportation Statistics (BTS) Office of Airline Information to establish which routes connect rural airports.[21]

[21] Rural airports are defined by the BTS as airports that had fewer than 100,000 commercial passengers departing from the airport by air in a year and satisfy at least one of the following conditions: i) the airport is not located within 75 miles of another airport from which 100,000 or more commercial passengers departed, ii) the airport was receiving essential air service subsidies as of August 5, 1997, or iii) the airport is not connected by paved roads to another airport.

To make sure that my results are not driven by the exposure to rural routes, I construct a measure of *Threat of entry* that excludes all routes connecting rural airports. I replicate my analysis using this alternative measure of threat and I report the corresponding coefficient in Table 11, Column 6. The effect of *Threat of entry* (excluding rural routes) on *Debt maturity* is 0.35 (while I get a coefficient of 0.36 when I use my main measure), ruling out the possibility that the result is driven by the exposure to rural routes.

5.4.7. Acela threat and debt maturity

To further support a causality argument, I consider the effect of the introduction of the Acela high-speed train on the debt maturity of airlines operating on affected routes. The Acela Express is Amtrak's fastest train in the United States, able to reach a peak speed of 150 mph (240 km/h). It connects Boston with Washington D.C. via 14 intermediate stops including Baltimore, Philadelphia, and New York City. Since its first ride on November 17, 2000, Acela has helped Amtrak to capture a 75% share of air/train commuters between New York and Washington (up from 37%) and 54% of the traffic from New York to Boston (up from 20%).[22]

High-speed trains are natural competitors for airlines over short to medium distances. Amtrak's fastest train makes the trip between Washington and New York in two hours 45 minutes, while planes travel the distance in one hour 20 minutes. Equivalent times for the New York-Boston trip are three hours 40 minutes by train, and one hour 15 minutes by plane. However, flights are often delayed because of weather or congestion. Amtrak arrives on time 90% or more of the time. According to transportation experts, when in-the-ground travel and waiting times are added Acela and air carriers' travel times are comparable.

Amtrak leased a high-speed train from Sweden for test runs in the Washington DC – New York City route in 1992. It is therefore unlikely that the decision to introduce high-speed trains was driven by the debt maturity change or the financial condition of airlines seven years later (Amtrak unveiled its plan for the Acela Express in March 1999.)[23] Therefore, I compute for each airline the exposure to Acela routes as the annual number of flights among airports in geographical proximity (i.e., I consider routes obtained by all pairwise combinations of the following cities: Newark (NJ), New York City, Boston, Philadelphia, Baltimore, Washington D.C.) over total annual flights. I consider as exposed the airlines having the 10% highest overlap in routes with Acela at the moment of the announcement. I consider all other airlines as non-exposed.

Hence, I regress *Debt maturity* on *Acela threat*, a dummy variable that takes a value of one for airlines highly exposed to Acela routes from the year of the announcement (1999) onwards.[24] Time and airline fixed effects are included and errors are clustered at the airline level. The estimated effect of the Acela threat on debt maturity is 0.21 (see Table 11, Column 7). The magnitude is twice that of the coefficient estimated from a comparable specification run for the threat of entry by low-cost airlines. There are, however, some conceptual differences between the two measures. In particular, different from the threat of entry, the announcement of Acela indicates entry in the following year with probability one. The stronger effect on debt maturity induced by the high-speed trains makes intuitive sense also because Acela's entry had arguably an even more disruptive effect on local airlines. Overall, results in this section provide suggestive evidence in support of the main result.

6. Threat of entry and debt composition

6.1. Credit supply: a discussion

Why should informed lenders extend long-term funding to threatened firms? When the borrower can pay back the full amount of the loan in expectation, risk-neutral lenders should be indifferent between long- and short-term debt. However, when a longer debt maturity diminishes the expected value of the repayment, the range of maturities available to the firm may be limited. In particular, risky borrowers could be excluded from the longer end of the maturity spectrum. In equilibrium, the resulting debt maturity will therefore be the one that maximizes the value of the firm conditional on the willingness of the lenders to offer debt at that maturity (Diamond, 1991a).

In my empirical tests, I cannot disentangle completely the role of the supply side. In line with the previous argument, both common intuition[25] and previous research (see, e.g., Brunnermeier and Oehmke, 2013) suggest that lenders should not impose an extension of debt maturity when the borrower is risky (as short-term debt is senior to long-term debt), thereby suggesting that my results are unlikely to be driven by the lenders.

The interactions between threatened airlines and lenders are however likely to affect debt composition. In particular, there are ex ante at least two reasons why threatened airlines should extend debt maturity mostly via loans instead of bonds. First, in the case of debt restructuring, the airline would benefit from having close ties to a specific lender (see, e.g., Diamond, 1991b; Hoshi et al., 1991; and Petersen and Rajan, 1994). Therefore, the debt contract should be renegotiated at a limited cost when the lender is unique. By contrast, a coordination problem prevents dispersed debt holders from easily renegotiating the debt contract (or would make it more costly), suggesting that it could be more difficult for the threatened airline to issue bonds in the first place.

Second, when entry is likely to trigger rollover failure, a firm may have an incentive to either undertake very risky projects—in an attempt to gamble for resurrection—or divert resources out of the firm before it defaults (for

[22] Data on traffic shares are reported in "Frustrations of air travel push passengers to Amtrak," *The New York Times* August 15, 2012.

[23] See "Amtrak unveils high-speed shuttle trains for busy travelers - Service between Boston and Washington is designed to compete with airlines." *Milwaukee Journal Sentinel* March 10, 1999.

[24] Airlines that were not operating in 1999 are excluded.

[25] See, for instance, "Greek Roll-over" posted on John Cochrane's Blog on June 3, 2015.

instance, by paying high dividends). A closer relationship between the financial institution and the borrower allows a better monitoring of the latter, ultimately reducing the agency costs (Park, 2000; Morrell, 2016). Overall, bond ownership is in most of the cases dispersed and usually denotes lose control on the issuer. Conversely, bank loans enforce tight control on the borrower (Brealey et al., 1977; Diamond, 1984; Fama, 1985), are usually senior (Welch, 1997), and collateralized (Rajan and Winton, 1995). An equilibrium that is optimal both for the firm and the lender, when the probability of entry is high, is more easily reached with a single lender (e.g., a bank or a large private lender), because monitoring and coordination costs are lower. Conversely, when the lenders are atomistic and dispersed (as bondholders are), it could become more difficult to design a debt contract that minimizes the inefficient costs deriving from a rollover failure.

Importantly, the arguments above hold under the assumption that repeated interactions with the lender decrease asymmetric information and renegotiation costs but do not allow the incumbent to refinance in bad times at no additional cost. If that were possible, the optimal strategy for the airline would be to keep maturity short and rely on its relationship with the lender in case of need. This arrangement is probably less common when the borrowers are small and risky airlines (which are arguably less important clients from a bank's perspective). However, it may contribute to explain why large, financially unconstrained airlines are less responsive to entry threats.

The next section provides some empirical evidence supporting the hypothesis that threatened airlines extend debt maturity mostly via loans.

6.2. Evidence from debt issuance

In this section, I explore how the threat of entry affects the maturity (in months) of the different debt issues. To conduct my analysis, I construct a sample of debt issues aggregating data from Mergent and Dealscan. Two important caveats are however in order. First, the data are incomplete. I only have data for debt issues reported in Mergent (mostly bonds) and Dealscan (mostly syndicated loans) that I am able to match to my initial sample. I do not have data on leasing contracts,[26] loans not included in Dealscan, and any other financial instruments not included in Dealscan or Mergent. Second, the maturity (in months) of each debt instrument is going to be a less effective proxy of rollover risk compared to the one used in the previous sections. Therefore, the analysis provided in this section has mostly the function of corroborating the evidence provided in Section 5.

My sample is highly dominated by bond issues. Out of 445 debt issues in my sample, 330 are bond issues and 115 are loans. Bonds display a significantly longer maturity on average (162 months versus 54 for loans) and a larger principal value. I run my regressions at the issue level including time but not firm fixed effects as issue data are sparse.

Table 12
Threat of entry and debt issuance.
This table presents results from regressions of *Maturity* (of each single debt issue, in months) on *Threat of entry* and controls. *Threat of entry* measures the percentage of routes under threat weighted by the number of paying passengers. The sample consists of debt issues collected from Dealscan and Mergent and matched by airline name to the main sample. Column 1 includes loans only, Column 2 includes bonds only, Column 3 includes both. *Relative size* is the size of the debt issue in dollar over book assets at the firm level. All other variables are described in Table A.1 in Appendix A. All regressions include an intercept (not reported) and year dummies. Standard errors are clustered at the airline level. *t*-statistics are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

	Maturity (in months)		
	Loans (1)	Bonds (2)	All (3)
Threat of entry	95.8026**	45.9176*	59.0525***
	(2.14)	(1.83)	(3.95)
Sales	−6.9866**	5.8095	10.0765*
	(−2.56)	(0.53)	(1.86)
Asset maturity	2.6538*	1.2574	5.3968***
	(2.02)	(0.54)	(4.63)
Tangibility	−30.0853	48.8194	−40.8623
	(−1.16)	(0.78)	(−0.96)
Profitability	110.4259	13.0961	27.2283
	(0.62)	(0.12)	(0.23)
ΔPassengers	118.6279	6.0764	7.9775
	(1.80)	(0.21)	(0.24)
Relative size	19.7563	0.3297	.5233***
	(0.87)	(1.39)	(3.37)
Asset backed		16.2290	
		(0.78)	
Redeemable		56.7119***	
		(3.10)	
Time fixed effects	Y	Y	Y
Observations	115	330	445
R-squared	0.501	0.337	0.292

My approach follows the related literature; see, e.g., Valta (2012).

An increase of one standard deviation in the threat of entry is associated with an increase of debt maturity of 10.5 months on average (see Table 12). The magnitude is however almost double for loans (Table 12, Column 1) than for bonds (Table 12, Column 2). This result contrasts with the fact that the average maturity in my sample is significantly longer for bonds, thereby suggesting that the access of threatened airlines to the bond market is somehow limited. This result is overall consistent with the hypothesis that agency and coordination costs make it easier for threatened airlines to extend debt maturity mostly via loans. This result is also consistent with the previous study by Denis and Mihov (2003), who suggest that high-quality firms issue bonds, while lower quality firms borrow from bank and non-bank private lenders. Overall, results in this section suggest that lenders are potentially aware of the threat of entry and the observed increase in debt maturity is the result of a bargaining process between airlines and banks.

7. Conclusion

The effect of the financial structure of firms on the competitive environment has been extensively debated.

[26] This seems however of secondary importance, as in unreported results I find that the threat of entry does not affect the percentage of aircraft leased.

Several papers have shown that leverage determines entry/exit decisions and output choices, and both empirical and theoretical work claims that the financial structure of firms influences how competition evolves. This paper posits that firms should, therefore, adapt their financial structure *before* competition increases, depending on what their expectation on future entry decisions is. More specifically, the question addressed by this paper is whether firms react to the threat of entry by disruptive competitors by increasing corporate debt maturity and/or decreasing leverage.

To give an answer to this question, I use market and financial data from the U.S. domestic airline industry. I propose an identification strategy based on the threat of competition posed by low-cost carriers' network expansion and exploit data on flight routes to build a measure of expected competition at the firm level. The largest part of the empirical literature on product market competition focuses on broadly defined industries to identify competing firms and generally assumes that all firms within the same industry are exposed to the same level of competition or to the same shocks. However, MacKay and Phillips (2005) show that the largest part of the variation in financial structure is explained by within-industry differences among firms. Airlines have some attractive features that make it possible to build a more precise identification strategy than similar studies based on all Compustat firms.

The main findings of this paper suggest that incumbents incorporate entry threats by low-cost competitors when making debt issuance decisions. Airlines anticipating tougher competition seek to increase the proportion of long-term liabilities while, at the same time, leaving their leverage ratio substantially unchanged. This result offers empirical confirmation of a growing series of theoretical papers deriving the optimal debt maturity structure in the presence of costly rollover risk (e.g., Diamond, 1991a; Brunnermeier and Yogo, 2009; and Diamond and He, 2014). Consistent with a rollover risk channel, most of the empirical findings reported in this paper are driven by financially constrained and risky airlines, i.e., those carriers that are less likely to refinance at a low cost if more efficient competitors enter the same market. Overall, my findings support the claim that the structure of the debt has important implications for the competitive environment and suggest that firms consider entry threats when making financing choices.

Appendix A

See Table A1.

Table A1
Description of variables.
This table provides a detailed description of the variables used. Airline information comes from Compustat, Compustat Industry Specific Annual, US Department of Transportation Form 41 filings, Dealscan, Mergent, and the T-100 Domestic Market database. Bond yields are from the Federal Reserve Board of Governors.

Variable	Definition
Sales	Logarithm of total sales
Size	Logarithm of total book assets
Total debt	Debt in current liabilities + Long-term debt
Profitability	Operating income before depreciation / Total assets
Tangibility	Net property, plant, and equipment (PPENT) / Total assets
Asset maturity	(Current assets/(Current assets+PPENT) × (Current assets/Cost of goods sold) + + (PPENT/(Current assets+PPENT)) × (PPENT/Depreciation and amortization)
ΔPassengers	Log change in passengers from year $t-1$ to t
Investment grade	Dummy = 1 if a firm is rated BBB- or better
ΔEmployees	Log change in employees from year $t-1$ to t
Chapter 11	Dummy = 1 if a firm files for Chapter 11 protection, data from Morrell (2016)
Investment	Capital expenditures / Total Assets
Q	(Total Assets + market value of equity - book equity)/ Total assets
Leverage	Total debt/Total assets
Debt maturity	Ratio of long-term debt (DLTT) - - debt maturing in two and three years (DD2+DD3) to total debt
Maturity5	Ratio of long-term debt (DLTT) - - debt maturing in two, three, four, and five years (DD2+DD3+DD4+DD5) to total debt
Term spread	The difference between the 10-year Treasury yield and the 3-month Treasury yield
Credit spread	The difference between BAA and AAA corporate bond yield
Leasing	Dollar leases over book assets
Long-term debt issuance	Newly issued long-term debt over book assets
Trend	Year-1990
Z-score	1.2 × (current assets - current liabilities)/total assets + 1.4 × (retained earnings/total assets) + 3.3 × (income/total assets) + 0.6 × (market capitalization/total liabilities) + + 0.9 (sales/total assets)
SA index	$-0.737 \times \text{Size} + 0.043 \times \text{Size}^2 - 0.040 \times \text{Age}$
Rural routes	# Routes connecting rural airports over total routes
Threat (excl. rural)	Percentage of routes under threat weighted by the number of paying passengers after rural routes have been excluded
Entry	Fraction of routes operated by an incumbent airline (weighted by passenger traffic) where at least one new low-cost airline enters for the first time
Avg. ticket price	Revenues per passenger over number of passengers carried

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